

miniPXI

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miniPXI: Development and Validation of an Eleven-Item Measure of the Player Experience Inventory

AQEEL HAIDER, KU Leuven, Belgium

CASPER HARTEVELD, Northeastern University, USA

DANIEL JOHNSON, Queensland University of Technology, Australia

MAX V. BIRK, Technologische Universiteit Eindhoven, Netherlands

REGAN L. MANDRYK, University of Saskatchewan, Canada

MAGY SEIF EL NASR, University of California, Santa Cruz, USA

LENNART E. NACKE, HCI Games Group, Games Institute & Stratford School of Interaction Design and Business, University of Waterloo, Canada

KATHRIN GERLING, KU Leuven, Belgium

VERO VANDEN ABEELE, KU Leuven, Belgium

Questionnaires are vital in games user research (GUR) to assess player experience (PX). However, having too many questions in surveys prevents wider uptake among GUR professionals because of games' rapid production cycles. To address this issue, we present the *miniPXI*—an eleven-item measure of the popular Player Experience Inventory (PXI)—providing single items for each of its eleven constructs. To develop the scale and examine its reliability and validity, we present three studies, conducted with 15 experts and 628 digital game players across continents. In the first survey study ($n = 366$, 15 experts), single items were selected. In a second survey study ($n = 232$), we explored reliability and validity of the single-item scale. Participants completed both full and single-item (SI) variants in three days. In the last study ($n = 30$), we established the validity and sensitivity via an experimental evaluation of two games. The results are nuanced; SI reliability estimates for PXI constructs range from .51 to .83 with an average of .68, we could confirm the validity for nine constructs. We conclude that the *miniPXI* can be a valuable tool for PX evaluations where a longer measure is not feasible, and provide practical considerations for its use.

CCS Concepts: • **Human-centered computing** → **HCI design and evaluation methods**.

Additional Key Words and Phrases: player experience, PX, games user research, GUR, single-item measure, scale, PXI, reliability, validation, questionnaire

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Authors' addresses: Aqeel Haider, KU Leuven, Belgium, aqeel.haider@kuleuven.be; Casper Harteveld, Northeastern University, USA, c.harteveld@northeastern.edu; Daniel Johnson, Queensland University of Technology, Australia, dm.johnson@qut.edu.au; Max V. Birk, Technologische Universiteit Eindhoven, Netherlands, m.v.birk@tue.nl; Regan L. Mandryk, University of Saskatchewan, Canada, regan@cs.usask.ca; Magy Seif El Nasr, University of California, Santa Cruz, USA, mseifeln@ucsc.edu; Lennart E. Nacke, HCI Games Group, Games Institute & Stratford School of Interaction Design and Business, University of Waterloo, Canada, lennart.nacke@acm.org; Kathrin Gerling, KU Leuven, Belgium, kathrin.gerling@kuleuven.be; Vero Vanden Abeele, KU Leuven, Belgium, vero.vandenabeele@kuleuven.be.

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1 INTRODUCTION

Games user researchers investigate whether player experience is delivered as intended by the game's designers [18]. For example, they evaluate whether players perceive a game level as sufficiently challenging or a cut scene as rousing curiosity. To this end, games user research (GUR) gathers empirical evidence through players' evaluations, and provides actionable insights for game development [13, 36]. In the past decade, GUR teams have increased in size and GUR is attracting growing attention in parallel with the ever-growing gaming market, which is forecasted to reach over \$250 billion in value by 2025 [55]. In addition, as games evolved beyond just entertainment aspects, understanding the user experience of games becomes increasingly crucial [13, 36]. The scope and significance of GUR must therefore be viewed not only through the lens of economic advantages, but also through the perspective of societal benefits. GUR methods have likewise evolved and expanded, ranging from playability heuristics [33, 62] to rapid iterative evaluations [69, 80], and from in-depth interviews and ethnographic field studies [18] to telemetry and biometric analysis [37, 72, 73]. However, because games are complex, interactive systems that aspire not only to be usable, but rather to provide an engaging experience, it remains hard to delineate how and when *successful* game experiences have been delivered. Moreover, despite its growing importance, GUR needs to subscribe to the rapid pacing and production cycles of game development [100, 112]. Therefore, questionnaires remain a core element of Games User Research—used widely in academic practices and by industry experts [21]. Typically, questionnaires are leveraged to assess player experience [31]. For example, to determine whether the overall player experience was enjoyable, whether players were given opportunity to experience competence, or whether the controls were usable on a basic level [17, 23, 56, 89].

Most recently, Abeele et al. [7] contributed the Player Experience Inventory (PXI), a validated player experience questionnaire that examines play experiences at the functional level (i.e., immediate experiences such as 'Ease of Control' or 'Clarity of Goals' that are a direct result of game design choices) as well as the psycho-social level (i.e., second-order emotional experiences such as 'Immersion' or 'Meaning'). In total, the questionnaire measures player experience by means of 10 constructs (not including the umbrella construct of 'Enjoyment') with a total of 30 items, which is comparable to the length of other player experience questionnaires such as the Player Experience Need Satisfaction questionnaire (PENS)[89], the Immersive Experience Questionnaire (IEQ) [56] or the Ubisoft Perceived Experience Questionnaire (UPEQ) [9].

Generally, such questionnaires that measure experiences through multiple dimensions and items are considered as providing higher reliability (precision and consistency of measurement) and validity (accuracy of the measurement), both cornerstones in psychometric measurement [76]. Yet, the length of player experience questionnaires can also be a barrier toward wider uptake [51]. Anecdotal accounts from game studios as well as requests from academic scholars highlight the need for a shorter version [48]. Moreover, in light of the widespread adoption of the System Usability Measure (SUS)—a 10-item small scale for rapid evaluation of perceived usability [19], the 11-item scale for analyzing player experience may prove useful for the GUR community. The need to keep the overall burden on the player as small as possible, leads to a desire to find short, but valid, instruments. This is particularly relevant considering the rapid iterations and the tight turnaround schedules that typify GUR in industry [100, 112], and when taking into account that academic research often deploys entire batteries of questionnaires, with player experience only being one of many aspects that are assessed [103]. Moreover, as more games than ever before are played on mobile phones [55], delivering questionnaires rapidly and in bite-sized format becomes increasingly important. Further, in a quest for rapid feedback from players, game studios sometimes embed single-item questions around central gameplay features into the game itself. For example, *Dota 2*

regularly asks players if they enjoyed a match, whereas *Fortnite* asks whether a weapon is over- or under-powered. Hence, shorter questionnaires are more suitable in many deployment settings, because they can be filled out more quickly by participants. Next to practical advantages, shorter questionnaires may also provide other advantages such as scores being easier to comprehend [35], and changes in scores may be more straightforward to interpret [107].

First efforts toward the creation of shortened variants of existing player experience questionnaires have been made. Most prominently, Harteveld et al. [48] investigated the potential of a shortened version of the PXI. Using only one item instead of three for each dimension, they evaluated an educational game used in engineering education. Their work suggests that with using 10 items, it is still possible to measure PX (based on factor analyses and a correlation with an enjoyment measure similar to the original work by Abeele et al. [7]). However, this preliminary study has several shortcomings to establish a shortened PXI for widespread use. It did not include a justification for the choice of specific items, nor investigate the reliability between the single item and the related dimension [109]. It was also limited in its specific context of use.

In our work, we aim to extend this preliminary work to develop an eleven-item variant of the Player Experience Inventory (PXI), i.e., a variant with one single item per construct, by systematically considering the reliability and validity of the included single items, thereby providing researchers and practitioners with a high-quality, yet efficient measure of player experience. To this end, the paper comprises three studies drawing input from experts and gamers across continents. In the first survey study, involving responses from 366 participants and input from 15 experts, single item for each of the PXI constructs were selected. We combined three different approaches (inspecting items mean, variance and normality, reliability with the constructs, ranking by experts, and legibility inspection). In a second survey study, based on a sample of 232 participants, reliability and validity of the eleven-item scale were verified. Participants completed both full and eleven-item variant in a time span of three days. In the last study, validity was again inspected as well as sensitivity, but this time in the context of an online experimental evaluation of two games. Here, 30 players were involved.

Our study presents a shorter eleven-item variant of the PXI, with an averaged reliability score obtained for the constructs at .68, in line with what has been proposed as threshold values in other disciplines (e.g., measures of quality of life [110] or job satisfaction [35, 74, 75, 107, 108]). However, our data also reveal challenges with respect to reliability and validity of specific constructs, in particular those that are more complex (e.g., Immersion and Mastery), and those that are immediate experiences (e.g. Progress Feedback or Ease of Control) when examined via delayed recall, suggesting that they should be subject to additional examination in future work.

Our work makes three key contributions: (1) We provide a validated eleven-item variant of the PXI questionnaire, the miniPXI, that enables a more efficient integration of the measure into existing GUR processes and academic research projects; (2) We discuss issues in reliability and validity of single-item measures of player experience on the basis of our work, outlining challenges to be addressed by future research; and (3) we provide actionable recommendations for researchers and practitioners to determine when to employ the eleven-item variant of the PXI, and when to use the longer version of the questionnaire.

2 BACKGROUND AND RELATED WORK

In the sections below, we first discuss the complex nature of player experience, and common GUR methods to assess it. Next, we focus on the use of questionnaires to evaluate player experience to fully situate the work presented here. Finally, we zoom in on the use single-item questionnaires to lay the methodological foundation for our work, exploring both psychometric as well as non-psychometric specificities.

2.1 Player Experience in Games

Precise and accurate data on players' perceptions of a game is pivotal for game user researchers: a thorough understanding of the experiences that a game offers to players is essential to verify whether the game designer's intent has been matched, and if not, to provide actionable recommendations as to how to achieve it. According to Mekler et al. [71], the central component of the player experience is game enjoyment, to be understood as "the positive cognitive and affective appraisal of the game experience, in part be associated with the support of player needs and values" (p. 927). However, in turn, to ensure that a player experience results in game enjoyment, different dimensions may need to be explored, such as flow [27, 28], immersion [20, 41], aesthetics [49, 52], or game usability measures [62].

In order to measure the multidimensional nature of player experiences, Games User Research (GUR) experts have a toolbox of methods available to them. In the lab, GUR professionals can make use of play and game usability testing [18], preferably in a rapid, iterative manner [70], combining observations of players with biometrics [72] and performance metrics. At a distance, game telemetry data can be captured [36], and drawing on game analytics [40], patterns can be found and communicated [105].

Either in the lab or at a distance, self-reports of players remain indispensable to provide insight into the underlying reasons of player experiences. Therefore, in complement to these methods, interviews and surveys are still used frequently [32] in GUR. Surveys can be particularly useful when collecting data at a larger scale [87] or from a distance. However, employing surveys appears to be deceptively quick and simple. Next to an appropriate sampling size and method, relevant questions need to be asked in an objective, reliable, and valid way [21]. Subsequent data cleaning and analysis necessitates a solid understanding of the statistical methods. Hence, the design and subsequent analysis of survey study is complex and error-prone [34, 81], challenging the utility for GUR. To this end, to assist GUR professionals, several questionnaires exist to aid in verifying the PX.

2.2 Questionnaires for Player Experience Evaluation

In the past years, different questionnaires have been developed and validated that can be used in PX evaluations, including: the Game Engagement Questionnaire (GEQ) measuring 4 different constructs (Absorption, Flow, Presence and Immersion) with 19 items [17]; Game Immersion Questionnaire (GIQ) with three subscales spanning seven constructs (Attraction, Time investment, Usability, Emotional attachment, Decreased perceptions, Presence and Empathy), totaling 24 items [23]; Immersive Experience Questionnaire (IEQ) with five constructs (Cognitive Involvement, Emotional Involvement, Real World Dissociation, Challenge and Control) totalling 31 items [56]; the Player Experience of Need Satisfaction questionnaire (PENS) with 5 constructs (Autonomy, Competence, Relatedness, Presence and Intuitive controls) [89] totaling 21 items or the Ubisoft Perceived Experience Questionnaire (UPEQ) measuring 3 constructs (Autonomy, Competence, Relatedness) with 21 items. Most recently, the Player Experience Inventory (PXI) [7] was proposed and validated. The evaluation of the full PXI with 529 participants showed a good model fit: χ^2/df of 2.089, CFI of .956 and a RMSEA of .045. Good convergent validity was evidenced by the composite reliability which was $\geq .7$ for all factors. Finally, discriminant validity was also good, as the square root of the average variance extracted was greater than any of the inter-construct correlations. The PXI is composed of two subscales: one subscale measures at the level of Functional Consequences (i.e., the immediate, tangible consequences) with five constructs (Ease of Control, Audiovisual Appeal, Challenge, Clarity of Goals, Progress Feedback), and the other at the level of Psychosocial Consequences (i.e., the emotional experiences) with 5 constructs (Meaning, Mastery,

Immersion, Autonomy, Curiosity). Additionally, game Enjoyment can be added, as a higher order umbrella construct, totalling 11 constructs and 33 items. The scale, through the measurement of PX at both the levels of game dynamics and game aesthetics, in line with the among game developers popular Mechanics-Dynamics-Aesthetics framework [54], has received recognition in providing actionable insight [14, 29] and has gained rapid traction [e.g. 8, 92, 113]. At the same time, the length of the scale may make it difficult to integrate in rapid, iterative game evaluations [69] or informal tests during events, or offered bitesized, via mobile phones. Researchers also found it difficult to integrate the scale in more comprehensive evaluations of serious games [11, 103]. As a result, preliminary attempts have been made to formulate a shorter variant with only one item per construct [48].

2.3 Single-item Measures in Empirical Science

Scales that rely on only one item to measure a construct are more generally known as *single-item measures* [109]. The practical relevance of using only one item rather than multiple items is obvious; the scale can be filled out more quickly and in this manner also more easily integrated in innovations cycles that rely on rapid iterative evaluations within tight schedules and budgets. However, there are also other, psychometric, advantages. Generally, a single-item scale is characterized by a higher face validity [35, 108]. Scores are easier to interpret, and can more easily be used to compare across different implementations. It has also been shown that there is less variability in adaptation across populations and contexts [74]. Moreover, shorter questionnaires also result in lower cognitive load, minimizing fatigue and frustration among participants, and consequently minimizing the amount of missing or invalid answers [38, 94]. As the number of items (and thus the length of the survey) decreases, respondents' willingness to devote time to completing and returning the questionnaire increases. Single-item measures' conciseness is therefore particularly advantageous for studies that are comprehensive and that require respondents to complete and return questionnaires autonomously.

However, single-item scales also present disadvantages. A first reason to oppose the use of single-item measures is that less information can be obtained by single-item measures than multiple-item measures [12]. This is particularly troublesome when constructs are complex and present inherent ambiguity [67]. A second argument against single-item questionnaires is the lack of an assessment of the measure's internal reliability. Because they allow for the calculation of correlations between items meant to measure the same construct, multiple-item scales are usually regarded as more trustworthy than single-item scales. If these correlations are positive and result in strong correlations (i.e., a high coefficient alpha), the scale is regarded to have internal consistency and to be a trustworthy measure of the construct under investigation. Moreover, the use of only one item also does not allow the measurement of explained versus unexplained variances which excludes the use of single-item measures in more advanced statistical models such as structural equation modeling. Consequently, some academic journals may consider multi-item measures as a must for acceptance of a manuscript for publication [106].

Despite the often assumed lower psychometric qualities, a number of studies have shown that in some instances, a single item is sufficient for assessing a specific idea or construct and even recommended for use [57]. Using a single item to measure a construct has various advantages as recommended by various researchers. Among these are increasing engagement in research at lower cost [109], less time required for filling in the questionnaire [75], less fatigue and boredom aroused due to redundant items [86], variability in adaptation across populations [75], and the simplicity with which it can be developed [44]. It is also possible to increase the face validity of a survey instrument when using single-item measures, particularly when the item is considered by respondents to be a direct measure of the construct under consideration [57]. Single-item measures'

conciseness is also likely to be advantageous for research projects that rely on respondents to complete and return questionnaires on paper or electronically, as opposed to surveys. Additionally using single-item is advantageous in terms of gaining broad design flexibility and item adaptability [57]. In one research based on changing the number of items, writers discovered that there might be more risk of participants turning to “mindless response behaviour” as the count of analogous items rises. They concluded that more items might not only take longer to calculate, but may also cause higher error rates in the results [38]. Single-item measures have been validated and applied in various fields such as quality of life for adult patients [110], medication adherence [58], social psychology [85], social behaviour [61], nursing research [111], social identification [83], job satisfaction [35, 74, 75, 107, 108], food quality [78], sports [64] and, perhaps most relevant for this study, e-commerce usability [24]. In this last case, authors found the single-item measure to provide adequate reliability and predictive validity, suggesting that in case of complex, multidimensional constructs such as usability [2], which is somewhat related to player experience, a single-item may still be useful.

The question then remains whether this also holds for the player experience. The aim of this study is therefore to develop and evaluate an eleven-item variant of the PXI and explore the reliability, validity, and sensitivity of this reduced scale.

3 PROCEDURE

To develop and validate the miniPXI, a three step process is applied as in Dolbier et al. [35]. To begin, single-item selection for each of the PXI constructs were made in Study 1, which included three sub-studies. In Study 1A, participants responded to a survey based on the current version of the PXI, these results were then used to verify items’ quantitative measures. In complement, in Study 1B interviews were conducted with 15 experts, including a range of game players (casual, hardcore, e-sports, and serious gamers), indie game developers, and GUR academics to rank and discuss items for each construct. In study 1C, a legibility analysis of the different items was performed. At the end of Study 1, these outcomes were combined and the item selection for the eleven-item scale was finalized.

In Study 2, the eleven-item scale’s reliability and validity were established via a survey study, in which participants were asked to rate a memorable play experience twice over the course of one week. The first survey required participants to complete the 33-item PXI. After three days, the same participants were invited to complete a second survey that included eleven-item PXI scale, along with selected items from the PENS [89] and AttrakDiff2 (AD) [50], to assess concurrent validity aspect of criterion validity.

Finally, in Study 3, to assess the eleven-item variant’s validity and sensitivity, an experimental study was conducted using a repeated measures design. Participants were asked to play 2 purposefully selected games and fill out the a single-item PXI, accompanied by selected items from PENS and AttrakDiff2. These scores were also compared to experts’ rankings of the games to establish expected differences between the games. Ethical approval was provided by a University Ethics Board for all three studies, with respectively numbers G-2020-2888-R3(AMD), G-2021-3869, and G-2020-2264-R3(AMD).

4 STUDY 1: SINGLE-ITEM SELECTION

This study comprised complementary approaches to select a single item for each PXI construct. First, an online survey was utilized to gather quantitative data with the goal of selecting single questions based on items’ normality, internal consistency, and discriminant validity. Secondly, structured interviews with game developers and designers were conducted to discuss the different items. Specifically, experts were asked to rank items, and next, to provide a brief explanation of

the reasoning behind their rankings. As a last step of Study 1, legibility of items was verified by computing reading ease scores and associate grade level. The R-based open-source statistical platform Jamovi version 1.6.23.0 [4], in combination with Microsoft Excel (Microsoft 365 Version 2203 Build 16.0.15028.20178), was utilized for statistical analysis. Below, we first detail the methods for each of these steps, we then present the combined results.

4.1 Method for Study 1

4.1.1 Study 1A: Survey Study of the 33-item Variant. An online survey containing the 33 items of the PXI was created. As prescribed, a 7-point Likert scale ranging from “Strongly Disagree (-3)” to “Strongly Agree (+3)” was used to score the PXI items for a recently played digital game, see Table 4. Additionally, the survey polled for participants’ age, gender, gaming habits, and level of gaming expertise. To avoid question order effects, all items were randomized following the recommended procedure by Malhotra [68] and two attention check items were added to control for invalid answers [79]. Links to the survey were circulated through social media, email lists, and students of the authors involved in the study. Data screening and cleaning were performed based on the procedure recommended by Gaskin and Carpenter [22, 45]. The response was dropped when a participant did not provide more than 10% of data or when suspicious answer patterns (graphic patterns or limited diversity in answers) indicated disengagement. Single-item reliability (SIR) for each construct was then calculated in three approaches.

The first method for estimating SIR is based on common factor analysis theory [24], using communalities h^2 of the single-item measure to provide an estimate of reliability, according to Harman [47]. Based on the recommended practices for EFA, items selected should have primary factor loadings $\geq .500$ with cross-loadings $\leq .300$ [15, 53], and selected items should load highly (> 0.6) with primary factors having adequate communalities ($> .5$).

A second method for computing SIR is using the procedure outlined in [47, 109], in which the SIR can be calculated using the well-known correction for attenuation formula suggested by Nunnally and Bernstein [77]. The correction for attenuation formula, used for this study, is as follows:

$$\hat{r}_{xy} = \frac{r_{xy}}{\sqrt{r_{xx} \cdot r_{yy}}} \quad (1)$$

where x and y represent the single-item and multi-item correlations, respectively. Variable $\hat{r}_{xy}$ is the assumed true correlation between x and y with the assumption that both were measured perfectly¹, r_{xy} is an observed correlation between x and y , r_{yy} is the reliability of the multi-item construct, and r_{xx} is the reliability of the single item. Knowing values for both r_{xy} and r_{yy} and the assumed attenuation value for $\hat{r}_{xy}$, the reliability of x can be calculated using this formula.

Finally, the SIR can also be evaluated by verifying internal consistency, i.e. by calculating Cronbach’s α [26] and McDonalds ω [39] for each construct, and Cronbach’s α if the item is dropped, to observe the effect of the individual item [84].

4.1.2 Method of study 1B: Structured interviews. In complement to the survey, online interviews with gamers/experts were conducted in a structured manner, starting with a ranking of items being used. Participants were recruited via the personal links of the authors of the paper; 15 gamers/experts agreed to participate. No further participants were invited as a sample of 15 is considered adequate to reach data saturation [16]. Individual construct definitions were presented, followed by three items that were used to measure that construct. Interviewees were asked to rank

¹As $\hat{r}_{xy}$ is assumed underlying correlation between perfectly measured x and y , the most conservative value for $\hat{r}_{xy} = 1.0$. As this assumption cannot be met in reality [90], and following the procedure in [24, 43, 109], an attenuated value is used for $\hat{r}_{xy}$ being 0.9.

items by dragging the preferred item to the top of the list, followed by the items that were less preferred. In addition, interviewees were requested to share a brief summary of their thoughts on the choices they made.

4.1.3 Method of study 1C: Legibility analysis. As a last step for the selection of single items, and to complement the results from the survey and to scientifically ground the discussion of legibility of the different items, the Flesch–Kincaid readability scores were computed to measure how understandable the items are to readers, based on their syntactical elements and style [60]. The readability scores consist of two sub-tests; the Flesch Reading-Ease (FRE), and the Flesch–Kincaid Grade (FKG) Level. The FRE rates text on a 100-point scale. The greater the score, the simpler it is to comprehend the text. The FKG calculates a number that relates to a U.S. grade level. It can also be interpreted as the number of years of education needed for someone to understand the text being assessed.

4.2 Results of Study 1

4.2.1 Results of study 1A: Survey study. Responses were received from 653 participants, of which 284 were dropped based on the two attention check items. After data screening and cleaning [22, 45], 3 more responses were dropped based on missing more than 10% of data, resulting in the final sample of 366 for analysis. Of the participants, 69.7% self-identified themselves as male ($n = 255$), 27% female ($n = 99$), 2.5% non-binary ($n = 9$), 2 preferred not to disclose and 1 preferred to self-describe. The mean age of participants was 24 years ($SD = 6.88$), ranging from 19 to 61 with a median score of 24. Participants were distributed geographically as follows: 49.5% from Europe ($n = 180$), 19.9% from North America ($n = 73$), 13.1% from Asia ($n = 48$), 12.8% from Oceania ($n = 47$), 4.6% from South America ($n = 17$), and 0.3% from Africa ($n = 1$).

First, Cronbach's α , and McDonald's ω were calculated to ensure the construct reliability (see Table 1) of the 11 constructs of the PXI, showing satisfactory internal reliability.

Then, single-item reliabilities for each item were evaluated using common factor analysis theory assessing communalities of the single-item measure to provide an estimate of reliability [24, 47]. In this step, it was also assessed whether the selected items are loading correctly and uniquely on their factors. Exploratory factor analysis was conducted, using the Principal Axis Factoring extraction method in combination with Promax rotation (see Table 2). Our sample was found adequate for factor analysis with a Kaiser-Meyer-Olkin (KMO) index of 0.839 and a significant Bartlett's test

Table 1. Reliability analysis (Cronbach's α , and McDonald's ω)

Construct	Cronbach's α	McDonald's ω
Audiovisual Appeal (AA)	.850	.855
Challenge (CH)	.801	.805
Ease of Control (EC)	.716	.734
Clarity of Goals (GR)	.716	.725
Progress Feedback (PF)	.739	.752
Autonomy (AUT)	.852	.857
Curiosity (CUR)	.874	.875
Immersion (IMM)	.689	.711
Mastery (MAS)	.742	.746
Meaning (MEA)	.843	.847
Enjoyment (ENJ)	.859	.862

Table 2. Factor analysis with 30 items, N = 366.

	AA	CUR	GR	MEA	AUT	CH	MAS	IMM	EC	PF	Communality (h^2)
AA_1	.774	.003	-.080	.035	.041	-.063	-.075	.033	.081	.052	0.62
AA_2	.836	.011	-.087	-.068	-.013	.076	.066	-.044	.002	.002	0.66
AA_3	.969	-.056	.043	-.001	-.011	-.017	-.042	-.081	-.076	-.033	0.77
CH_1	.006	-.006	-.059	.001	.035	.802	-.035	-.021	.014	-.023	0.60
CH_2	-.025	.030	-.027	.015	-.038	.866	-.066	-.022	.048	.043	0.72
CH_3	.006	-.017	.118	-.012	.008	.621	.077	.111	-.118	.041	0.50
EC_1	-.051	.019	.030	.056	-.012	-.055	-.007	.025	.802	.031	0.62
EC_2	.031	-.009	.218	-.066	.028	-.036	.028	.036	.632	-.027	0.59
EC_3	.057	.015	.056	-.002	.002	.155	.061	-.004	.391	-.046	0.32
GR_1	.054	-.020	.543	.045	-.051	.014	-.032	.001	-.002	.112	0.38
GR_2	-.053	.020	.796	.017	-.084	-.020	-.102	-.065	.146	-.040	0.58
GR_3	-.072	-.058	.762	.045	.089	.020	-.046	.004	-.004	-.075	0.50
PF_1	.012	-.020	-.027	-.015	-.034	.022	.070	.025	-.002	.875	0.78
PF_2	.000	-.032	.434	-.116	.036	-.036	.105	-.047	.031	.250	0.390
PF_3	.001	.081	.185	.071	.025	.026	-.053	-.006	-.016	.625	0.57
AUT_1	-.004	-.016	-.133	.095	.824	.015	.014	.017	.063	-.001	0.73
AUT_2	-.016	.025	.071	-.098	.728	-.036	.021	.001	-.013	-.001	0.52
AUT_3	.041	.014	.029	.055	.873	.043	-.059	-.023	-.038	-.015	0.80
CUR_1	-.009	.768	-.177	.077	.026	-.052	.097	-.012	-.001	.071	0.68
CUR_2	-.066	.986	.041	-.119	-.009	.067	-.021	-.041	.001	-.038	0.80
CUR_3	.038	.817	-.003	.024	.019	-.012	-.100	.017	.034	.023	0.70
IMM_1	-.108	-.062	-.229	-.022	-.019	.100	-.093	.914	.073	.078	0.65
IMM_2	.183	.119	.193	.124	-.048	-.062	.013	.425	-.075	-.043	0.51
IMM_3	.001	.015	.144	-.058	.039	-.076	.109	.636	-.013	-.074	0.51
MAS_1	-.030	-.031	-.233	.041	-.056	-.049	.823	.020	.130	.063	0.57
MAS_2	.058	.047	.094	.027	-.010	.117	.520	-.043	.024	-.110	0.45
MAS_3	-.048	-.022	.088	-.005	.061	-.044	.793	-.015	-.112	.037	0.60
MEA_1	.007	-.044	-.031	.876	-.019	.000	.021	.016	-.014	.056	0.75
MEA_2	-.043	.077	.156	.674	-.040	.014	.055	-.083	-.063	-.087	0.51
MEA_3	.003	-.085	-.079	.874	.080	-.004	-.018	.017	.104	.065	0.74

Principal axis factoring' extraction method was used in combination with Promax rotation. Based on the full PXI scale development process, extraction factors were fixed to 10. Selected items should load highly (> 0.6) with primary factors having adequate communalities ($> .5$). For clarity, items with primary factor loadings above $\geq .500$ are in black, selected items are in bold.

of sphericity ($\chi^2(435) = 4922, p < .001$). The highest value was for MEA_1 (.876) with $h^2 = .753$ and lowest value was for IMM_3 (.636) with $h^2 = .567$. Then, means, standard deviations, item-rest correlations, Cronbach's α , and McDonald's ω if item dropped were computed as well as the SIR via the attenuation formula for each item at $\hat{r}_{xy} = .9$ respectively, see Table 3.

4.2.2 Results of study 1B: Interviews. Among the 15 interviewed gamers/experts, they self-identified as esports expert ($n = 4$), dedicated gamer ($n = 4$), indie game developer ($n = 3$), serious game player ($n = 2$), and casual gamer ($n = 2$). Participants were geographically distributed across Europe ($n = 6$), Oceania ($n = 5$), and North America ($n = 4$). Ranking scores were provided and are shown in Table 3

4.2.3 Results of study 1C: Legibility. Finally, legibility was assessed followed by a verification of the SIR scores, the FRE scores, and the FKG Level respectively. FRE scores varied between 54.7 and

Table 3. Means, standard deviations, item-rest correlations, Cronbach's α , and McDonald's ω if item dropped, SIR for each item calculated at $\hat{r}_{xy} = 1$ and .9 respectively, FRE score, FKG level, and Interview rankings. Selected items are bolded.

	Mean	SD	Item-rest correlation	If item dropped		Reliability (r_{xx})		Flesch Tests		Interview Rankings		
				Cronbach's α	McDonald's ω	$\hat{r}_{xy} = 1$	$\hat{r}_{xy} = .9$	FRE Score	FKG Level	First	Second	Third
AA_1	2.20	0.99	.71	.80	.82	.59	.72	100	0.8	1	7	7
AA_2	2.27	0.89	.72	.80	.80	.61	.76	100	0	11	2	2
AA_3	2.09	1.11	.75	.77	.77	.66	.82	54.7	7.3	3	6	6
CH_1	1.33	1.36	.65	.72	.72	.53	.66	100	1.8	2	8	5
CH_2	1.41	1.33	.70	.67	.67	.61	.75	71.8	5.2	2	5	8
CH_3	1.53	1.31	.59	.79	.79	.44	.54	71.7	6.7	11	2	2
EC_1	1.78	1.27	.58	.57	.58	.47	.58	88.9	3.8	5	6	4
EC_2	2.10	1.05	.58	.60	.60	.46	.57	95.1	2.4	4	6	5
EC_3	1.53	1.37	.47	.72	.73	.31	.38	92.9	2.2	6	3	6
GR_1	2.23	1.01	.49	.69	.70	.34	.41	92.9	2.2	0	6	9
GR_2	2.32	0.97	.57	.59	.60	.45	.55	100	0	11	3	1
GR_3	2.44	0.83	.56	.61	.61	.44	.55	66.7	5.6	4	6	5
PF_1	1.82	1.26	.64	.56	.56	.56	.69	95.1	2.4	1	8	6
PF_2	1.83	1.18	.45	.78	.78	.27	.34	72.6	5.8	5	4	6
PF_3	1.66	1.30	.61	.60	.60	.50	.62	87.9	3.7	9	3	3
AUT_1	1.83	1.25	.75	.77	.77	.66	.81	100	0.5	12	2	1
AUT_2	1.75	1.33	.65	.86	.86	.50	.62	89.8	4.1	0	8	7
AUT_3	1.63	1.35	.77	.75	.75	.70	.86	89.8	4.1	3	5	7
CUR_1	1.60	1.37	.73	.85	.85	.61	.75	82.3	3.7	5	6	4
CUR_2	1.63	1.39	.78	.80	.80	.70	.86	94.3	2.3	3	4	8
CUR_3	1.52	1.50	.77	.82	.82	.67	.83	66.1	6.2	7	5	3
IMM_1	0.68	1.71	.52	.62	.62	.40	.49	74.8	5.8	3	4	8
IMM_2	1.80	1.12	.49	.63	.66	.35	.43	100	0.5	10	2	3
IMM_3	1.74	1.22	.56	.54	.57	.46	.56	90.9	2.3	2	9	4
MAS_1	1.58	1.24	.59	.63	.64	.47	.58	100	1	9	3	3
MAS_2	1.80	1.09	.51	.72	.72	.35	.43	78.8	3.9	3	8	4
MAS_3	1.57	1.25	.61	.61	.61	.50	.62	84.9	3.6	3	4	8
MEA_1	1.30	1.35	.75	.74	.74	.67	.83	78.8	3.9	13	2	0
MEA_2	1.31	1.37	.64	.85	.85	.48	.59	87.9	2.4	1	4	10
MEA_3	1.25	1.47	.74	.75	.75	.65	.81	78.8	3.9	1	9	5
ENJ_1	2.42	0.79	.77	.77	.78	.68	.85	100	0.5	4	9	2
ENJ_2	2.43	0.75	.70	.83	.84	.57	.70	54.7	6.6	0	4	11
ENJ_3	2.29	0.88	.74	.80	.80	.64	.79	100	0.8	11	2	2

100. FKG levels scored between level 6.7 and 0 (lowest level). FRE scores and FKG levels are shown in Table 3.

4.2.4 Final selection of single-items. Combining the information from the three studies, a final selection was made of the single items, based on expert rankings first, next verifying the reliability scores and, finally, verifying the legibility, see Table 4.

For **Audiovisual Appeal**, the item “I liked the look and feel of the game” (AA_2) was selected, based on strong support from experts ($n = 11$) and a SIR score of .76. Even though the item “I appreciated the aesthetics of the game” (AA_3) had the highest SIR (.82), it was not considered due to low legibility scores. For **Challenge**, the item “The game was not too easy and not too hard to play” (CH_1) was retained, with a SIR score of .66 and highest legibility. The items “The challenges in the game were at the right level of difficulty for me” (CH_3) and “The game was challenging but not too challenging” (CH_2) was dropped due to low SIR score (.54) and low legibility scores, respectively. For **Ease of Control**, the item “It was easy to know how to perform actions in the game” (EC_1) was selected, with the highest SIR and legibility score. Although, the item “I thought the game was easy to control” (EC_3) was ranked highest by experts, it was dropped due to a very low SIR score (.38). For **Clarity of goals**, the item “The goals of the game were clear to me” (GR_2)

Table 4. 33-item PXI with selected single-item per construct in bold

Functional Constructs			Psychosocial Constructs		
Construct	Item Code	Item	Construct	Item Code	Item
Audiovisual Appeal	AA_1	I enjoyed the way the game was styled	Autonomy	AUT_1	I felt free to play the game in my own way
	AA_2	I liked the look and feel of the game		AUT_2	I felt like I had choices regarding how I wanted to play this game
	AA_3	I appreciated the aesthetics of the game		AUT_3	I felt a sense of freedom about how I wanted to play this game
Challenge	CH_1	The game was not too easy and not too hard to play	Curiosity	CUR_1	I wanted to explore how the game evolved
	CH_2	The game was challenging but not too challenging		CUR_2	I wanted to find out how the game progressed
	CH_3	The challenges in the game were at the right level of difficulty for me		CUR_3	I felt eager to discover how the game continued
Ease of Control	EC_1	It was easy to know how to perform actions in the game	Immersion	IMM_1	I was no longer aware of my surroundings while I was playing
	EC_2	The actions to control the game were clear to me		IMM_2	I was immersed in the game
	EC_3	I thought the game was easy to control		IMM_3	I was fully focused on the game
Clarity of Goals	GR_1	I grasped the overall goal of the game	Mastery	MAS_1	I felt I was good at playing this game
	GR_2	The goals of the game were clear to me		MAS_2	I felt capable while playing the game
	GR_3	I understood the objectives of the game		MAS_3	I felt a sense of mastery playing this game
Progress Feedback	PF_1	The game informed me of my progress in the game	Meaning	MEA_1	Playing the game was meaningful to me
	PF_2	I could easily assess how I was performing in the game		MEA_2	The game felt relevant to me
	PF_3	The game gave clear feedback on my progress towards the goals		MEA_3	Playing this game was valuable to me
Enjoyment	ENJ_1	I liked playing the game			
	ENJ_2	The game was entertaining			
	ENJ_3	I had a good time playing this game			

was maintained, with the greatest SIR and legibility scores, as well as the highest rankings from 11 experts. For **Progress Feedback**, the item “*The game gave clear feedback on my progress towards the goals*” (PF_3) was retained, ranked highest by 11 experts with SIR score of .67 and high legibility score. Although the SIR score was slightly higher (.69) for the item “The game informed me of my progress in the game” (PF_1), it was dropped due to low ranking by experts. For **Autonomy**, the item “*I felt free to play the game in my own way*” (AUT_1) was selected, with a SIR score of .81 and the highest legibility score, it was also ranked first by 12 experts. Although the SIR score was slightly higher (.86) for the item “I felt a sense of freedom about how I wanted to play this game” (AUT_3), it was ranked first by only 3 experts. For **Curiosity**, the item “*I wanted to explore how the game evolved*” (CUR_1) was retained. It was ranked first by 5 experts, has a SIR score of .75 and adequate legibility scores. Even though the item “I felt eager to discover how the game continued” (CUR_3) had a SIR score of .83 and was ranked highest by experts ($n = 7$), the word “eager” was believed to be a bit odd by experts, and was reflected in the low legibility scores. The item “I wanted to find out how the game progressed” (CUR_2), despite having highest SIR score (.86), was not considered due to the low ranking from experts. For **Immersion**, the item “I was fully focused on the game” (IMM_3) was selected, with SIR score of .56 and satisfactory legibility score. Although the item “I was immersed in the game” (IMM_2) was ranked highest by 10 experts, it was dropped due to a low SIR score (.43). For **Mastery**, the item “*I felt I was good at playing this game*” (MAS_1) was selected, being ranked highest by experts ($n = 9$) with a SIR score of .58 and the highest legibility scores. The item “I felt a sense of mastery playing this game” (MAS_3), despite having highest SIR score (.63), was not considered due to the low ranking by experts. For **Meaning**, the item “*Playing the game was meaningful to me*” (MEA_1) was retained, having a SIR score of .83 with being ranked highest by experts ($n = 13$) and high legibility scores as well. The items “Playing this game was valuable to me” (MEA_3) and “The game felt relevant to me” (MEA_2) were not considered due to low expert ranking and low SIR score (.59) respectively. Finally, for **Enjoyment**, the item “*I had a good time playing this game*” (ENJ_3) was selected, with a SIR score of .79 and ranked highest by experts ($n = 11$). Even though the SIR score for the item “I liked playing the game” (ENJ_1) was higher (.85), it was dropped due to low rankings from experts.

5 STUDY 2: ASSESSING RELIABILITY AND VALIDITY OF THE MINIPXI THROUGH A SURVEY STUDY

The aim of the second study was to verify the reliability and validity of the single items, selected in Study 1, via a procedure akin to a test-retest procedure, in which participants were asked to rate a memorable play experience twice over the course of one week.

5.1 Method of Study 2

5.1.1 Procedure. Two online surveys were created on Qualtrics [3], with the first survey similar to the one used in Study 1A, containing the 33-item PXI, age, gender, gaming platform, and level of gaming expertise as well as two attention check items. At the end of the first survey, participants were also asked to fill out their email addresses. Three days after responding to the first survey, participants were invited to complete the second survey, using customized URLs created from their email addresses. A second reminder was sent 2 days later. The second survey contained only the selected PXI single items from Study 1 and a attention check items. In addition, to verify concurrent validity, items from conceptually related constructs from the PENS [89] and AttrakDiff2 [50] were added, identical to the procedure in [7]. Both of these scales have been validated and are widely utilized in the community. For the validation of original PXI [7], the same scales and matching constructs were utilized. Therefore, the same procedure was used for the miniPXI validity analysis.

In both surveys, questionnaire items were randomized to reduce response bias due to the ordering. Distribution of the flyer and survey link took place through social media marketing, email lists, and distribution via the networks of the co-authors of the study. To encourage the participation and completion of the second survey, a raffle was conducted to distribute 5 Amazon gift cards worth \$50 each.

5.1.2 Data analysis. Similar to Study 1A, Jamovi version 1.6.23.0 [4] was used for most statistical analyses. Additionally, R version 4.0.5 [99] was used for data pre-processing. The data analysis process was divided into three steps. In the first step, internal consistencies (Cronbach's α , and McDonald's ω) were calculated for all PXI constructs and selected constructs from PENS and AttrakDiff2. Data collected in the first survey was then used to calculate internal consistencies of PXI constructs as the second survey only had a single item per PXI construct. In the next step, the reliability of every single item was measured by performing bivariate correlation analysis between the single item and the mean score of the full scale following the procedure put forward by Dolbier et al. [35]. To determine the degree to which the replies to the same item varied across parts 1 and 2, the correlation scores of each single-item with itself were also calculated, see Table 5. Finally, the single items' concurrent validity was determined using bivariate correlation analysis with the mean scores of conceptually related constructs from PENS and AttrakDiff2 using a procedure similar to the full PXI scale validity analysis [7], see Table 6.

5.2 Results of Study 2

5.2.1 Participants. A total of 851 responses were received for the first survey, of which 287 were removed due to the fact that they did not complete the entire survey. Another 89 replies were removed after they failed to correctly respond to the two randomly induced attention check items, identical to those used in Study 1A. From the 475 respondents who completed the first survey, 300 responded to the second survey, with three respondents dropping out because they did not complete the full survey and 53 dropping out because of the attention check items, resulting in a sample of 244 respondents.

Next, following the same procedures as in Study 1A, data screening and cleaning were carried out, resulting in removal of 6 responses. In addition, 2 duplicate responses and 4 responses based on mismatched emails from parts 1 and 2 were deleted. This resulted in a final sample of 232 responses for data analysis.

Of the participants, 152 were experienced players (more than 5 hours of gameplay per week), 66 were casual players (1-5 hours of gameplay per week) and 14 were non-active players (less than 1 hours of gameplay per week). In this sample, 72.4% self-identified themselves as male ($n = 168$), 27.2% female ($n = 63$) and 1 non-binary. The mean age of participants was 25.2 years ($SD = 7.12$),

Table 5. Means, standard deviations, internal consistencies (Cronbach's α and McDonald's ω) for full scale and correlations of single item with full scale and with item itself.

	Full Scale (3-items per construct)				Single Item			
	Mean	SD	Internal Consistency		Mean	SD	Correlation	
			Cronbach's α	McDonald's ω			3-item	Same Item
Audiovisual Appeal	2.28	0.77	.82	.83	2.28	0.73	.61***	.51***
Challenge	1.45	1.05	.76	.76	1.48	1.21	.62***	.58***
Ease of Control	1.94	0.86	.67	.71	1.93	1.00	.46***	.36***
Clarity of Goals	2.40	0.72	.82	.83	2.22	0.93	.58***	.45***
Progress Feedback	1.89	0.87	.64	.67	1.84	1.02	.54***	.48***
Autonomy	1.84	1.09	.82	.82	1.81	1.17	.64***	.58***
Curiosity	1.73	1.16	.87	.87	1.85	1.20	.69***	.64***
Immersion	1.59	1.02	.61	.65	1.88	1.17	.62***	.60***
Mastery	1.92	0.82	.68	.69	1.96	0.89	.54***	.55***
Meaning	1.57	1.15	.81	.82	1.54	1.24	.68***	.61***
Enjoyment	2.45	0.65	.80	.80	2.30	0.84	.61***	.54***

*** Significant at the 0.001 level (two-tailed).

ranging from 18 to 55 with a median score of 24. Participants were distributed geographically as follows: 48.2% from Europe ($n = 111$), 37.5% from North America ($n = 87$), 9.6% from Oceania ($n = 22$), 3.4% from Asia ($n = 10$), and 0.9% from South America ($n = 2$).

5.2.2 Reliability. Means and standard deviations for full scale and single items, internal consistency of PXI constructs (Cronbach's α and McDonald's ω), and bivariate correlation scores are detailed in Table 5. The multiple item PXI constructs of *Challenge*, *Meaning*, *Autonomy*, *Clarity of Goals*, *Audiovisual Appeal* and *Curiosity* demonstrated high internal consistency ($\alpha = .76-.87$, $\omega = .76-.87$). Constructs of *Immersion*, *Progress Feedback*, *Ease of Control* and *Mastery* showed moderate reliability ($\alpha = .61-.68$, $\omega = .66-.69$). Additionally, overall *Enjoyment* ($M = 2.45$, $SD = 0.65$), demonstrated high reliability ($\alpha = .80$, $\omega = .80$). For related multiple item constructs from PENS and AttrakDiff2 (AD), *AD Pragmatic Quality*, *PENS Autonomy*, *PENS Intuitive controls*, *PENS Competence* and *AD Attractiveness* demonstrated high internal consistency ($\alpha = .72-.84$, $\omega = .73-.86$). The construct of *PENS Presence*, however, showed moderate reliability ($\alpha = .68$, $\omega = .68$).

From bivariate correlation analysis, single items of all PXI constructs, other than *Ease of Control* ($r = .46$, $p < .001$), were found to be strongly correlated ($r > .5$, $p < .001$) with their associated multiple item PXI construct, see Table 5. The highest score was observed for the construct of *Curiosity* ($r = .69$, $p < .001$). The correlation scores of single items with themselves (across timepoints) were surprisingly low ($r < .5$) for the the construct of *Ease of Control*, *Clarity of Goals*, and *Progress Feedback*. Highest and lowest scores were observed for the construct of *Curiosity* ($r < .64$) and *Ease of Control* ($r < .36$), respectively.

5.2.3 Validity. Criterion validity of the single items was partially (concurrent validity) assessed by correlating them with conceptually related constructs from the PENS and AttrakDiff2 scales, see Table 6. There were significant positive correlations ($r > .55$, $p < .001$) observed for the constructs of *Autonomy*, *Ease of Control*, *Immersion*, *Curiosity*, and *Mastery*. However, only moderate correlations were observed for the constructs of *Challenge*, *Audiovisual Appeal*, and *Meaning* ($r = .34-.38$, $p < .001$). Surprisingly, the most striking results to emerge from this analysis were the correlation scores for the constructs of *Clarity of Goals* ($r = .09$, $p = .098$) and *Progress Feedback*

Table 6. Study 2: Concurrent validity of the PXI single items

Construct PXI	Construct PENS or AD	Pearson's r
Audiovisual Appeal	AD Attractiveness	.34***
Challenge	PENS Competence	.38***
Ease of Control	PENS Intuitive controls	.63***
Clarity of Goals	AD Pragmatic Quality	.09
Progress Feedback	AD Pragmatic Quality	.07
Autonomy	PENS Autonomy	.69***
Curiosity	PENS Presence	.56***
Immersion	PENS Presence	.58***
Mastery	PENS Competence	.56***
Meaning	AD Attractiveness	.34***

*** Significant at the .001 level (one-tailed).

($r = .07, p = .076$). These low scoring results will be revisited in the discussion section to provide reflections and possible explanations for the lower scores.

6 STUDY 3: ASSESSING VALIDITY AND SENSITIVITY OF ELEVEN-ITEM PXI THROUGH AN EXPERIMENTAL STUDY

The goal of this last study was to verify the validity of the eleven-item PXI for an active game experience, and additionally in exploratory manner assess the sensitivity of single items. Validity was assessed by comparing single-item per construct with the related constructs from PENS and AttrakDiff2 for two different games. Sensitivity of each of the items was assessed by comparing differences found between two different games, to the differences expected by experts, as well as differences found by the multiple-items of the related PENS and AttrakDiff2 constructs.

6.1 Method of Study 3

6.1.1 Procedure. The overall study was designed as a within-subject (repeated measures) experiment. All gaming sessions were virtually supervised using Zoom. Qualtrics [3] was used to create and administer the online survey. Participants were first asked to fill out demographic information along with information on their gaming platform. They were then asked to play each of the two online games, counterbalanced, for at least 6 minutes and not more than 10 minutes. After playing each game, participants were asked to respond to randomized questions, including PXI single items and items from related constructs of PENS and AttrakDiff2, probing for their experience from the game they just played. At the end of the study, they were given the opportunity to ask questions related to the study and research, and thanked for their participation. Each participant was compensated with a \$10 Amazon gift card at the end of the session

6.1.2 Games selection. The games were selected via Itch.io, a website for browser-based indie games. Different games were assessed on feasibility and gameplay. Based on this, two browser-based online games (*Red Handed* (A) [82] and *Evoland* (B) [1]) were shortlisted following the guidelines for selecting games for research projects [102]. *Red Handed* is an action shooting stealth game, mixed with some puzzle elements, where players have to get avoid being caught. *Evoland* is a 2D platform adventure game. Players are able to unlock additional gameplay mechanics, gameplay systems, and graphics as they progress through the game's story-line, in this manner also being provided a history of computer games.

6.1.3 Expert heuristics. Next, four game researchers who are not involved in the study, but active as scholars in GUR, were asked to rate the games on each of the PXI constructs, on a scale from 1 (low construct score) to 5 (high construct score), to establish heuristics with respect to how the different games differ on the different PXI constructs. Therefore, definitions of the constructs and a brief explanation of the process were all that were provided to these four researchers. These ranking scores of four experts were then averaged for each game and per construct. The difference between the rankings then served as a heuristic for sensitivity analysis, i.e. the greater the difference in experts' rankings, the greater the expected difference for the single-item PXI.

6.1.4 Participants. Power and effect size calculation ($\delta = 0.5$, $\alpha = 0.05$, power = 0.8, one-tailed) suggested that at least 27 participants would be required, assuming medium effect sizes [25]. Considering some dropouts, 34 participants were recruited through word of mouth, snowball sampling, mailing lists and social media sites. Four responses were dropped based on wrong answers on randomly induced attention check items, leaving the final sample of 30 responses for further analysis.

Of the 30 participants, 14 were experienced players (more than 5 hours of gameplay per week), 10 were casual players (1-5 hours of gameplay per week) and 6 were non-active players (less than 1 hours of gameplay per week). In this sample, 60% self-identified themselves as male ($n = 18$) and 40% as female ($n = 12$). The mean age of participants was 26.9 years ($SD = 5.02$), ranging from 20 to 40 with a median score of 27. On average, sessions lasted about 40 minutes, including the 2 gaming sessions and completing the questionnaire twice. Participants were distributed geographically as follows: 60% from North America ($n = 18$), 23.2% from Europe ($n = 7$), 10% from Oceania ($n = 3$), and 6.6% from Asia ($n = 2$).

6.1.5 Data analysis. Jamovi version 1.6.23.0 [4] was used for quantitative data analysis. Pre-processing of raw data was done in R version 4.0.5 [99]. To match the scoring scheme used for different scales (i.e. PENS, AttrakDiff2 and PXI), data collected from PXI was mapped from a scale of -3 to +3 to a scale of 1 to 7. The data analysis process was divided into two steps. In the first step, the single items' concurrent validity, for an active game experience, was determined using the same procedure as Study 2. In the next step, paired-samples t-tests with Games (*Red Handed*, *Evoland*) as within-subject factors were calculated, with Bonferroni correction, for all PXI constructs and selected constructs from PENS and AttrakDiff2. We further supply Cohen's d as a measure of effect size.

6.2 Results of Study 3

Means, standard deviations, the average ranking for each game and rating difference per construct, t score, significance (one-tailed), and effect size (Cohen's d) for each construct of PXI and related constructs from PENS, and AttrakDiff2 are detailed in Table 8.

6.2.1 Validity of the miniPXI in an experimental setting. Criterion validity of the single items, for an active game experience, was assessed by correlating them with conceptually related constructs from the PENS and AttrakDiff2 scales, see Table 7. For both games, significant positive correlations ($r > .5$, $p < .001$) were observed for all PXI constructs except for *Clarity of Goals*, and *Progress Feedback*. Moderate degree of correlation were observed for the construct of *Meaning* in both game A ($r = .366$, $p < .05$) and game B ($r = .396$, $p < .05$). Similarly, the scores for the construct of *Clarity of Goals* were significant for game A ($r = .372$, $p < .05$) but were not significant for game B ($r = .233$, $p = .216$). Consistent to previous findings, it was observed that constructs of *Progress Feedback* had lower correlation scores and were not significant for both game A ($r = -.024$, $p = .900$) and game B ($r = .251$, $p = .180$).

Table 7. Study 3: Concurrent validity for an active game experience with Red Handed (game A) and Evoland (game B)

PXI Construct	PENS and AD	Pearson's r (A)	Pearson's r (B)
Audiovisual Appeal	AD Attractiveness	.61***	.68***
Challenge	PENS Competence	.72***	.71***
Ease of Control	PENS Intuitive controls	.68***	.56***
Clarity of Goals	AD Prgmatic Quality	.37*	.23
Progress Feedback	AD Prgmatic Quality	-.02	.25
Autonomy	PENS Autonomy	.75***	.50***
Curiosity	PENS Presence	.79***	.65***
Immersion	PENS Presence	.77***	.81***
Mastery	PENS Competence	.91***	.78***
Meaning	AD Attractiveness	.36*	.40*

* $p < .05$, ** $p < .01$, *** $p < .001$ (one-tailed)

Table 8. Study 3: Means, SD, the average ranking for each game and rating difference per construct, t score, significance (one-tailed), and effect size (Cohen's d) for each construct of PXI and related constructs from PENS, and AtrakDiff2.

		Red Handed (A)			Evoland (B)			Ranking	Paired Sample t-test (A-B)			
		Mean	SD	Ranking	Mean	SD	Ranking	Diff. (A-B)	t	df	p	d
PXI	Audiovisual Appeal (AA)	4.67	1.54	2.75	4.90	1.63	3.50	-0.75	-0.712	29	0.482	.130
	Challenge (CH)	4.97	1.52	3.75	4.80	1.61	3.25	0.50	0.424	29	0.674	.078
	Ease of Control (EC)	5.03	1.52	3.25	4.80	1.58	3.25	0.00	0.592	29	0.559	.108
	Clarity of Goals (GR)	5.87	1.41	3.50	5.17	1.62	2.25	1.25	2.041	29	0.050	.373
	Progress Feedback (PF)	5.17	1.44	3.50	5.00	1.60	3.75	-0.25	0.431	29	0.670	.079
	Autonomy (AUT)	4.63	1.90	3.25	4.30	2.04	3.50	-0.25	0.639	29	0.528	.117
	Curiosity (CUR)	2.37	1.59	3.50	3.37	1.50	4.75	-1.25	-3.525	29	0.001	.644
	Immersion (IMM)	2.17	1.39	3.00	2.87	1.59	4.25	-1.25	-2.911	29	0.007	.532
	Mastery (MAS)	4.60	1.92	3.00	4.40	1.57	2.75	0.25	0.541	29	0.592	.099
	Meaning (MEA)	3.43	1.61	2.75	3.70	1.75	4.00	-1.25	-0.757	29	0.455	.138
	Enjoyment (ENJ)	5.10	1.40	3.75	5.03	2.03	3.50	0.25	0.166	29	0.869	.030
PENS	Competence	4.64	1.68		4.45	1.39		0.55	0.55	29	0.587	.100
	Presence	3.05	1.08		3.52	1.20		-2.271	-2.271	29	0.031	.415
	Intuitive Control	4.92	1.19		5.32	1.28		-1.243	-1.243	29	0.224	.227
	Autonomy	4.20	1.41		4.04	1.44		0.487	0.487	29	0.630	.089
AttrakDiff2	Attractiveness	4.80	1.17		4.99	1.11		-0.663	-0.663	29	0.512	.121
	Pragmatic Quality	4.43	1.00		4.68	0.97		-0.942	-0.942	29	0.354	.172

6.2.2 Sensitivity of the miniPXI. Results of the sensitivity analysis are in line with the heuristics provided by the experts, and also confirm the sensitivity of the related constructs of the PENS and AttrakDiff2. Noteworthy, where experts expressed a clear difference (>1), a paired sample T-test they revealed a significant differences as well. Following experts' heuristic for the construct of *Curiosity*, we found significant difference ($t(29) = -3.525, p = .001$) between game A ($M = 2.37, SD = 1.59$) and game B ($M = 3.37, SD = 1.50$). Similarly, as expected, the difference was significant ($t(29) = -2.911, p = .007$) for the construct of *Immersion*, between game A ($M = 2.17, SD = 1.39$) and game B ($M = 2.87, SD = 1.59$). The difference was also found significant ($t(29) = -2.271, p = .031$) for the PENS *Presence*, conceptually related with *Curiosity* and *Immersion*. The established heuristic also holds true for the construct of *Clarity of Goals*, as the difference between game A ($M = 5.87, SD = 1.41$) and game B ($M = 5.17, SD = 1.62$) was significant ($t(29) = 2.041, p = .050$).

7 DISCUSSION

The aim of this paper was to explore the feasibility of, shortened, eleven-item variant of the Player Experience Inventory by examining the reliability, validity and sensitivity for a single item for each of the different constructs. In the sections below, findings are discussed with respect to these psychometric aspects. Additionally, the implications that our findings hold for the evaluation of player experience in general are given, as well as practical considerations for the use of the eleven-item variant of the PXI.

7.1 Reliability, Validity and Sensitivity of the miniPXI

Here, we summarize our key findings with respect to reliability, validity and sensitivity of the eleven-item variant of the PXI, and discuss possible explanation for differences in performance between the shortened and full version of the questionnaire.

7.1.1 Summary of questionnaire performance. With respect to reliability, validity and sensitivity of the eleven-item PXI, our findings draw a mixed picture. Regarding *reliability* (i.e., the consistency of a measurement), there are several aspects that warrant further discussion. In our work, multiple approaches toward the assessment of reliability were taken as recommended in [24, 109]. Both factor analysis and the correction for attenuation calculation showed reliability estimates for the PXI constructs that range from a low of .51 to a high of .83 with an average across constructs of .67 (communalities) and .68 (SIR). On average, these values are similar to the values found by Wanous et al. [109] and deemed acceptable; for a single-item measure, reliability scores close to .70 are considered reasonable. As a whole, this suggests that the single-item variant of the PXI yields a satisfactory reliability score. However, upon a closer examination of reliability estimates for the different constructs, five out of eleven constructs (Progress Feedback, Clarity of Goals, Ease of Control, Immersion, Mastery) range between .5 and .6. The first three are at the functional level, the latter are at the psycho-social level. These scores are lower than what is commonly recommended.

With respect to *validity* (i.e., the accuracy of measurement), our study confirmed positive associations between PXI constructs and related constructs of the PENS and AttrakDiff2 questionnaires, which aligns with findings for the original version of the PXI [7]. However, there are two exceptions: the constructs of “Progress Feedback”, and to some extent, “Clarity of Goals”. Given the aforementioned concerns regarding the reliability of these two items, issues with validity are not be surprising [77, 98].

With respect to the *sensitivity* analysis (i.e., the capability of detecting differences between measurements), we find that the eleven-item variant performs well. Comparing the effect sizes, the eleven-items perform similar to the related multi-item constructs from the PENS and AttrakDiff. Additionally, the constructs that are predicted by experts as different are also flagged as significant by the single items. This suggests that despite the earlier lower reliability and validity of some constructs in Study 4 and Study 5, the single-item constructs perform equally well in discriminating different player experiences from each other as the more elaborate, multiple item constructs.

7.1.2 Discussion of shortcomings. One aspect that may contribute to lower reliability estimates observed here is the *original low number of items per construct* in the full variant of the PXI. The full version of the scale is already quite conservative in length, including three items per construct only. It is in the nature of the calculation of internal reliability and thus the computation of the single-item reliability that fewer original items lead to a lower value when further reduced to one; the calculation of Cronbach’s α is also affected by the length of the test [77, 98]. In contrast, in most of the existing studies that validate single-item measures and that have guided our work with respect to threshold values for acceptable reliability, the number of items included in the

original measure were higher (≥ 5 per construct) [24, 35, 46, 107]. Additionally, there is still no standard threshold for player experience evaluations; as suggested by Lance et al. [65], the decision of setting a threshold for accepted reliability values depends mostly on the purpose of the study and what type of information a researcher is particularly interested in. We note that thus far, in PX evaluations, there are no other studies of single-item variants that can guide us in determining what is an acceptable threshold. We hypothesize that a lower single-item reliability estimate may particularly applicable to those constructs that are known to be complex and multidimensional, e.g. Immersion and Mastery. This is discussed in the next section on the multi-dimensionality of player experience.

Lower reliability scores may also be the *result of the specific method* chosen in Study 1 and Study 2, i.e. a survey study where participants were asked to evaluate a game they played in the past, i.e. delayed recall. Hence, lower correlations may be attributed to the effect that this delayed recall can have on the accuracy of reporting of player experiences [91]. In particular, it may be that as a player experience is no longer in active memory, recalling experiences correctly, particularly at the level of functional consequences (the immediate, direct experience of game mechanics), may be hampered. Our findings lend support to this hypothesis; the design of Study 2 (Section 2) is akin to a test-retest setup. Here, we measured delayed recall of a recent game experience, measured via two surveys that are three to maximally five days apart, with the exact item being asked across both surveys, once as part of 33 PXI items, and once as part of 11 items. Observed bi-variate correlations between responses on the same item were moderate (see Table 5, last columns); where one would expect to see a stronger correlation (other meta-analytic studies typically present correlations above .6, see [109]), we observed scores ranging from .36 to .64. Particularly, for the three constructs at the level of functional consequences (Ease of Control, Clarity of Goals and Progress Feedback) we found scores below .5. Additional confirmation of this hypothesis can be found in the stronger validity scores of Study 3, as compared to Study 2. In Study 3, players were given an active game experience and asked to immediately evaluate this. Moreover, in the validation of the full PXI, Abeele et al. [7] also observed this difference in validity scores between delayed and immediate recall.

In summary, it remains difficult to assess to what extent lower reliability scores of the single-item variant are acceptable in the case of player experience or, whether they are a result of the inherent multidimensional nature of some of the constructs, or whether they point toward a problematic item or a problem in the delayed recall method, or any combination, pointing toward multiple avenues that future work should explore to increase rigor in player experience evaluation.

7.2 Implications of Our Findings for Player Experience Research

Our findings also hold implications for player experience research more widely. Here, we discuss what the complexity of PX constructs means for their examination by means of questionnaires, and we address the impact of temporal factors on immediate or recalled play experiences.

7.2.1 Addressing the multidimensionality of PX constructs. Our results show that the reliability of the constructs of *Immersion* and *Mastery*, both at the psycho-social level, scored lower than other constructs at this level, included in the single-item PXI. Interestingly, these two constructs have previously been studied in relation to player experience, and are recognized as multidimensional constructs. *Immersion* has been promoted as central to game enjoyment by several academic games researchers. Brown and Cairns [20] distinguish different levels of immersion, ranging from engagement and engrossment to ultimately full immersion. Ermi and Mayra [41] differentiate between sensory, challenge-based, and imaginative immersion, hinting at the different aspects of games that can contribute to it. Several researchers also equate immersion to 'presence', or a sense of 'being in a virtual space' [42, 66, 97]. The complexity of the construct of Immersion is

also reflected in other measures, e.g., the Immersive Experience Questionnaire [30, 56], which contains no less than 31 items across six different dimensions that measure immersion, focusing on Basic attention, Temporal dissociation, Transportation, Challenge, Emotional involvement and Enjoyment. A similar analysis can be made for the construct of *Mastery*, which bears resemblance to related constructs of Competence [88] in Self-Determination Theory or self-efficacy beliefs [10]. Yet, it is also often linked to experiences of Flow [27], which refers to an optimal game experience [95, 96]. When deconstructing Mastery against the backdrop of Flow, it becomes clear that the construct includes multiple dimensions such as the balance between player skill and challenge presented by the game, but also the merging of action and awareness. Taking into account previous research demonstrating that single-item scales perform worse when underlying psychological constructs are complex or inherently ambiguous [67, 109], we believe that this issue may also apply to particularly these two constructs of player experience. Hence, we conclude that when immersion or mastery are key focus points of evaluation, it is advised to use measures that accurately reflect the multi-dimensionality and assist in their deconstruction; single-item measures may not be the right choice. Nevertheless, single-item measurements may still be good enough to generally screen for these concepts.

7.2.2 Exploring how temporal aspects impact perceived PX. Our results also show that the reliability of the constructs of *Clarity of Goals*, *Ease of Control* and *Progress Feedback*, all three at the functional level, score lower than is recommended. Here we hypothesize that in particular the method of delayed recall may have had an impact on stability of recall. However, our studies show distinct differences in self-reporting of player experience depending on the time of reporting, i.e. whether it was evaluated immediately after play, or if recall took place at a later time. Here, studies on the conceptually related construct of user experience [63, 104] explicitly acknowledge the temporal element in the perception of experiences, with definitions of user experience such as ISO 9241-210 [6] encompassing “a person’s perceptions and responses that result from the use or anticipated use of a product, system or service [...] including all the users’ emotions, beliefs, preferences, perceptions, physical and psychological responses, behaviors and accomplishments that occur *before, during, and after* use”, clearly embracing that perceptions of the same interactive technology may fluctuate over time [59]. While a similar argument could be made for player experience, we are not aware of test-retest studies that examine how player perspectives on PX-related constructs change over time, e.g. how *Challenge* or *Meaning* is experienced during, immediately after, or several days post-play. In our work, we contribute a first insight into the impact of time on perceived PX. Our findings suggest that the perception of player experience itself may not be stable, particularly for constructs of the level of functional consequences, and fluctuate substantially, even when measured only a few days apart. With respect to the validity analysis, stronger correlations were observed in Study 3 (experimental study with immediate recall) than in Study 2 (survey study with delayed recall).

7.3 Practical considerations for the Use of the miniPXI

In our work, the eleven-item PXI performed the strongest when used in an experimental setup, where the participants evaluated the same games. Therefore, we suggest that it can be a valuable instrument and a feasible alternative for multi-item measures in the context of evaluating active experiences of a specific game or game prototype within a larger evaluation. As previously discussed, single-item measures also have practical advantages, including faster response time, cost effectiveness, high face validity, decreased respondent frustration, and less missing data [74, 106], suggesting that there is a trade-off that games researchers can make when deciding which variant of the questionnaire to employ. Specifically, the eleven-item PXI may be preferable in situations where the PXI is one among many measurement instruments, or in situations where a longer measure would not be

feasible at all (e.g., when evaluating player experience in time-constrained field research). However, we caution against the use of the single-item per construct instrument in situations that rely on delayed recall, or where player experience is compared across different games, genres and audiences, or where in particular Mastery or Immersion is to be examined. Here, a multi-item per construct measure is recommended in order to achieve reliable measurements. Using the multiple-item measure instead of the single-item one has the most apparent advantage of greater reliability and validity. Hence, we conclude that preference is given to a multi-item questionnaire where possible and the examination of constructs related to immersion of mastery be at the core of the study.

7.4 Limitations and Future Work

There are several open questions that invite further exploration. A first investigation is whether different single items, perhaps not part of the original PXI would show a higher reliability. In this aspect, when reviewing the PX literature, we noticed that there is no established culture that addresses the issue of test-retest reliability of measures of player experience, and the stability of the construct over time. Here, we see an opportunity for future work that examines in detail player experience that is reported during, immediately after, and with a time delay post-play. The reliability estimates obtained in this study can serve as a useful benchmark. Along the same lines, we also recommend further examination of the constructs of *Progress Feedback* and *Clarity of Goals*, which both performed below average. It should be reviewed as to whether they are sufficiently distinct to warrant two different constructs.

Second, this study depended on the selection of a single item from the existing items in the original PXI rather than the development of a new item for each underlying construct. This may account for the low reliability scores found for a few of the constructs. In addition, few of the available studies on scale development have examined a newly designed single-item [24, 35]. It would be an intriguing future study to test a newly generated single-item per construct and then compare it to the miniPXI to determine whether it improves the low reliability scores, particularly for the constructs with low reliability values.

Thirdly, Study 1 and Study 2 relied on delayed recall of a gaming experience. This differs from how gaming experiences are typically evaluated by GUR practitioners. Our results suggest that studies on delayed recall provide lower validity scores, and this is echoed by other researchers. Nevertheless, it might be interesting to understand more thoroughly how the design of the study (delayed recall versus active experience, one game versus a pool of games) impacts findings with respect to player experience.

Finally, a different pathway toward the provision of a shortened version of the PXI—which is the overarching research goal of our work—could be to reduce the number of considered constructs (rather than limiting the number of items per construct). In fact, several researchers have deployed the original PXI using only a few of the constructs [5, 52, 93, 101]. This would likewise limit demands placed on respondents through reduction of questionnaire items. We believe that this is an interesting option because the single-item variant of the PXI still contains eleven items for eleven constructs, yet holding one umbrella construct “Enjoyment”. This is also related to the fact that human experience is intricate and unique, and that there is a great deal of room for error in measurement, particularly when evaluations rely on self-report methods, which in our case are questionnaires. Therefore, in order to obtain accurate evaluations, “triangulations” consisting of multiple items and multiple scales are required to be explored.

8 CONCLUSION

Questionnaires are a common means of assessing player experience both within industry and academia. From a psychometric perspective, questionnaires that measure experiences through

multiple dimensions and items are preferred, as generally, they provide higher reliability (precision and consistency of measurement) and validity (accuracy of the measurement), both cornerstones in psychometric measurement. However, the length of the currently available player experience questionnaires can also act as a barrier towards wider uptake, considering the rapid iterations and the tight turnaround schedules that typify GUR in industry, and the deployment of multiple test batteries in academic settings. Single-item measures have many practical advantages, including faster participant response times, cost effectiveness, high face validity, decreased respondent frustration, and fewer missing data.

This leads to a desire to find shorter, but valid, instruments. In our work presented here, we focus on the development of the eleven-item variant of the Player Experience Inventory [7], i.e. a variant where only one item is used for each of its eleven constructs, thereby providing a drastically shortened questionnaire. Based on three studies carried out with 15 experts and a total of 628 participants, we contribute with the miniPXI, a shortened version of the original questionnaire. However, our results regarding reliability and validity suggest that the shortened version of the questionnaire does not perform at the same level as the full version for all constructs, with complex ones in particular scoring lower. This suggests that there is a trade-off between the quality of the measurement instrument and its length, and that games researchers need to carefully weigh which variant of the questionnaire to employ. For example, in situations where the PXI is only one among many measurement instruments, or in situations where a longer measure would be inappropriate (e.g., when evaluating player experience in time-constrained field research), the miniPXI may be advised. However, in light of its limitations, preference should be given to a multi-item per construct questionnaire in situations that rely on delayed recall, or where player experience is compared across different games, genres and audiences. Here, using the multiple-item measure instead of the single-item one has the most apparent advantage of greater reliability.

Finally, our work also highlights the complex, multidimensional and temporal nature of player experience and calls for further studies to unpack where methodological limitations surface, and where researchers face the rich yet elusive nature of player experience.

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